

Lab 8

Clustering Techniques Implemented using WEKA and Python:

1. DBSCAN (Density-Based Clustering)
2. . ROCK (Robust Clustering using Links)

Dataset : [Mall Customer Segmentation Dataset](#)

- **WEKA**

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Clusterer output
=== Run information ===

Scheme:      weka.clusterers.DBSCAN -E 0.9 -M 6 -A "weka.core.EuclideanDistance -R first-last"
Relation:    store_customers-weka.filters.unsupervised.attribute.Remove-weka.filters.unsupervised.attribute.Remove-R1
Instances:   1000
Attributes:  4
  Gender
  Age
  Annual Income (k$)
  Spending Score (1-100)
Test mode:   evaluate on training data

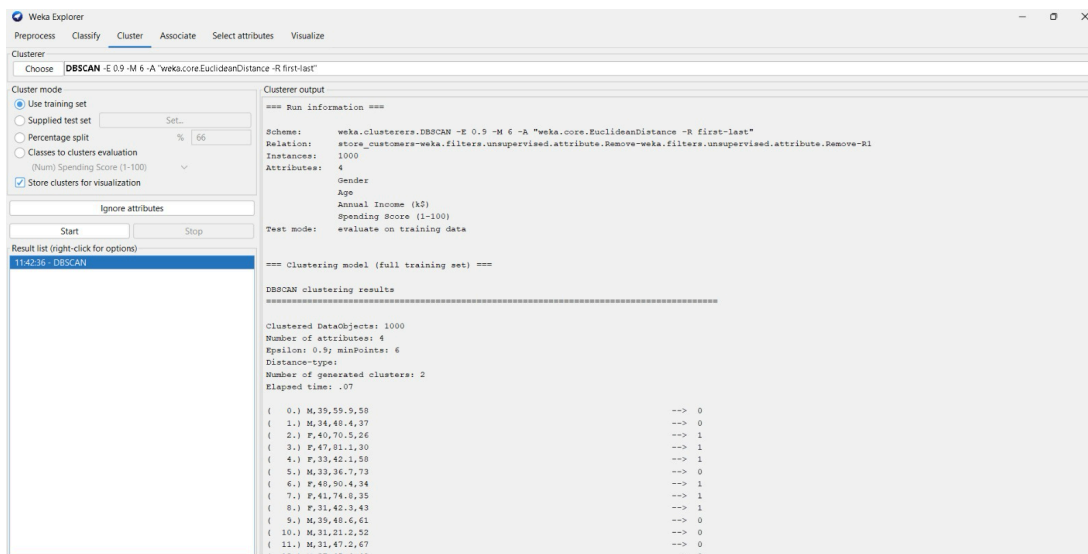
=== Clustering model (full training set) ===

DBSCAN clustering results
=====

Clustered DataObjects: 1000
Number of attributes: 4
Epsilon: 0.9; minPoints: 6
Distance-type:
Number of generated clusters: 2
Elapsed time: .07

( 0.) M,39,59.9,58      --> 0
( 1.) M,34,48.4,37      --> 0
( 2.) F,40,70.5,26      --> 1
( 3.) F,47,81.1,30      --> 1
( 4.) F,33,42.1,58      --> 1
( 5.) M,33,36.7,73      --> 0
( 6.) F,48,50.4,34      --> 1
( 7.) F,41,74.8,35      --> 1
( 8.) F,31,42.3,43      --> 1
( 9.) M,39,48.6,61      --> 0
(10.) M,31,21.2,52      --> 0
(11.) M,31,47.2,67      --> 0
(12.) M,31,47.2,67      --> 0
(13.) M,31,47.2,67      --> 0
(14.) M,31,47.2,67      --> 0
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(92.) M,31,47.2,67      --> 0
(93.) M,31,47.2,67      --> 0
(94.) M,31,47.2,67      --> 0
(95.) M,31,47.2,67      --> 0
(96.) M,31,47.2,67      --> 0
(97.) M,31,47.2,67      --> 0
(98.) M,31,47.2,67      --> 0
(99.) M,31,47.2,67      --> 0
(100.) M,31,47.2,67      --> 0
```

Fig. 8.1: DBSCAN clustering



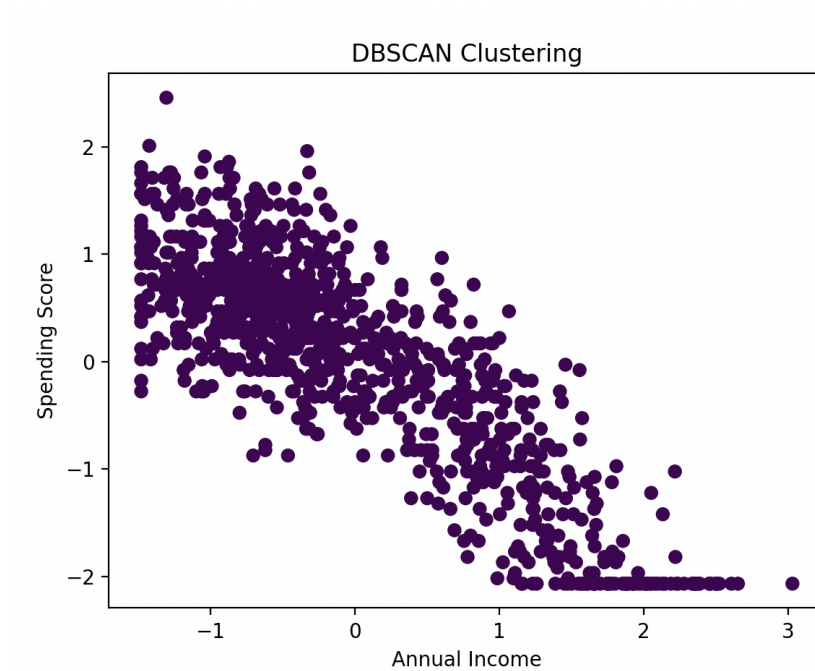


Fig.8.3: DBSCAN clustering

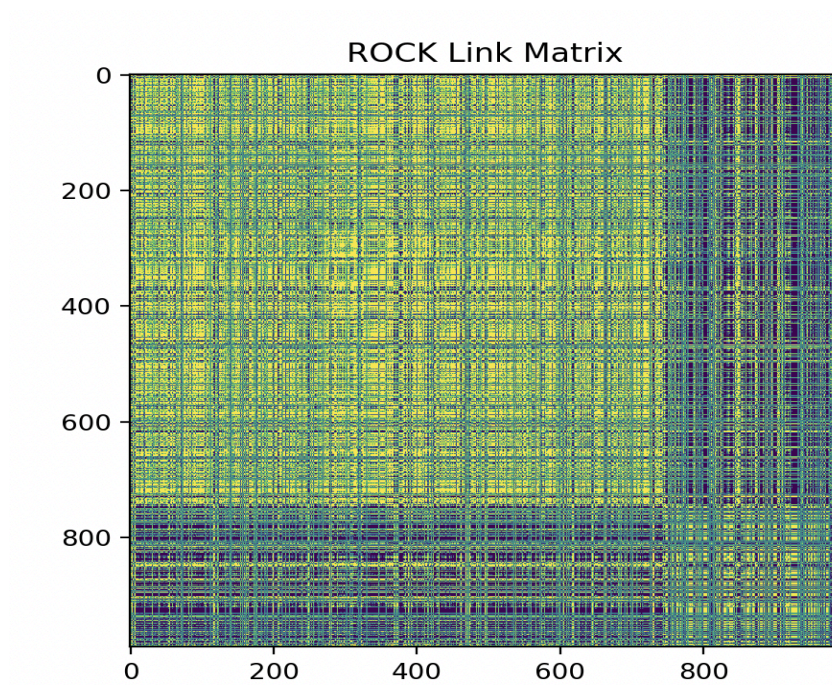


Fig.8.4: ROCK clustering